Check the robot. Tune the shooter. Build an auto path.

Robot: one of last season's practice robots. Field: tape on a wall and on the floor.

Goal

Leave with checked drive and turn numbers, a launcher speed table we measured ourselves, and one autonomous path that ends inside a taped parking box.

Safety

Stand behind the robot whenever the launcher can spin. One student drives; a different student keeps a hand near **Stop** and may press it at any time. Press Stop and wait for the wheel to stop before loading, measuring or touching the robot. Clear a jam only with the robot switched off.

Plan

Time	What we do
0:00-0:10	Goal, safety, first jobs: driver, Stop, recorder, measurer. Swap often.
0:10-0:50	One pair checks the robot's numbers: battery, distance, turns, launcher (page 2). Everyone else builds the practice field (page 3).
0:50-1:35	Check the camera, then tune the shooter at three distances (page 4).
1:35-1:45	Break. Press Stop first.
1:45-2:45	Build the auto path and run it empty (page 5).
2:45-3:00	Type the numbers into the code, save, photograph the tape layout.

Controls

Program	Controls
Tune Drive	INIT: D-pad UP/DOWN picks 24 in forward/back; LEFT/RIGHT picks a 90-degree turn. Start runs it once.
Tune Shooter	INIT: D-pad LEFT then DOWN picks RED AUDIENCE. After Start: sticks drive slowly; hold A to face the tags; D-pad UP/DOWN changes speed by 25; hold right bumper to spin; tap Y for one short shot once the screen says At speed.
Autonomous Route	INIT: D-pad LEFT then DOWN picks RED AUDIENCE; A picks Start A; X switches dry run on/off. A dry run never spins or feeds.

One robot runs at a time. Everyone else tapes the route, graphs the results, or reads the code for the next step.

Robot checks worksheet

Do these first: the shooting marks and the route assume the robot drives the distance it is told. Take the balls out.

Battery (INIT, any program): volts _____ GOOD / SWAP / CHARGE (circle). SWAP or CHARGE: change it before going on.

Which way the Hub faces (Tune Drive, INIT only)

Logo faces _____ USB faces _____ Match LOGO / USB in Heading.java? yes / fixed. Quarter turn right by hand: heading _____ to _____ (should go UP about 90). Lift front / lift one side: heading moved _____ / _____ (only a few degrees). Both pass: show the mentor, who sets MOUNTING_CONFIRMED = true.

Distance: ticks per inch (Tune Drive)

Line a mark on the robot up with a tape mark. Run it, write the ticks and seconds from the screen, press Stop, then measure with a tape. **Ticks per inch = average of left and right ticks / real inches.** The code starts at about 45. If your average is more than 1 away, type it on your robot's TICKS_PER_INCH_... line near the top of robot/Drive.java (for last season's goBILDA: TICKS_PER_INCH_PRACTICE_GOBILDA) and run once more.

Asked for	Left ticks	Right ticks	Measured (in)	Seconds	Ticks per inch
24 in forward					
24 in forward					
24 in backward					
Average					
New number: 24 in forward					

Turns (Tune Drive)

Line the robot up with a taped square corner, then run each turn.

Asked for	Degrees off (by eye)	Seconds	Wobbled at the end?
90 degrees right			
90 degrees left			

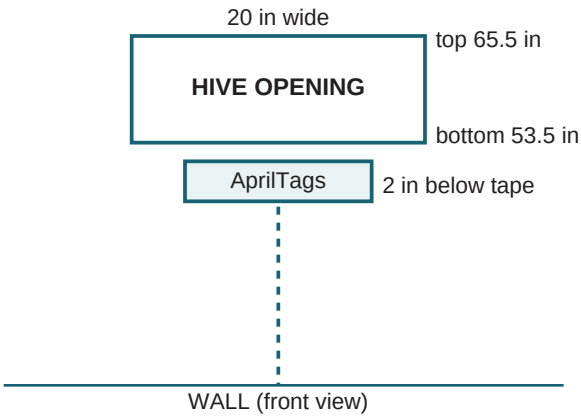
Launcher speed (Tune Shooter, no balls)

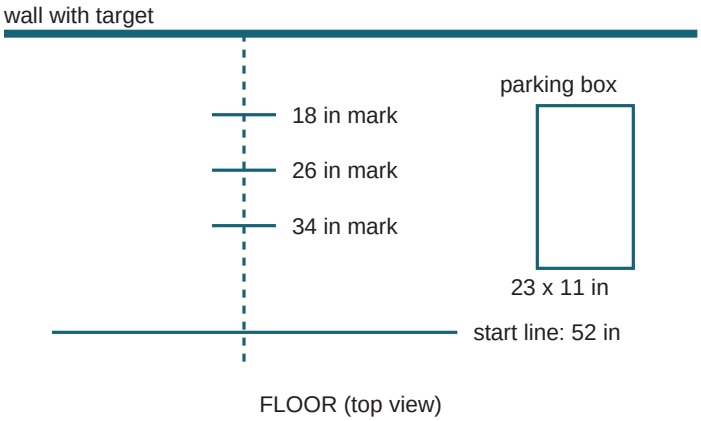
Start, hold the right bumper, count the seconds until At speed: true, write the measured speed once it settles. Then D-pad UP eight times (200 more) and repeat. Never reaching the speed asked for: check the battery, then tell the mentor.

Speed asked for	Speed measured	Seconds to At speed
Normal: _____		
Normal + 200: _____		

Build the practice field

The wall stands in for our hive. The sizes come from the game manual.





Tape this	Measurement	It stands for
Wall rectangle	20 in wide. Bottom edge 53.5 in above the floor, top edge 65.5 in.	The upward CELL opening, seen from in front.
AprilTags	Centered under the rectangle. Top of the black squares 2 in below the bottom tape.	The tags under the CELL.
Center line	Straight out from the wall, under the middle of the rectangle.	Lining up on our hive.
Start line	Across the center line, 52 in from the wall.	The field wall we start against (our estimate).
Shooting marks	Front bumper 18, 26 and 34 in from the wall.	The room between hive and field wall.
Parking box	23 in by 11 in, off to one side.	The LOADING ZONE at its real size.

AprilTag sheets

Target (choose in INIT)	Tag IDs	Packet pages: right / left
BLUE AUDIENCE	38, 39, 40, 41	6 / 7
BLUE SCORING (far side)	42, 43, 44, 45	8 / 9
RED AUDIENCE - use this one	34, 35, 36, 37	**10 / 11**
RED SCORING (far side)	30, 31, 32, 33	12 / 13

Print at Actual Size / 100%. A black square measures **3.25 in**. Tape the LEFT sheet to the left of the RIGHT sheet so the printed center marks meet; the labels go below the tags. Tilt the webcam up until INIT shows **Tag seen: true** from the 34-inch mark. Use last season's balls. If the launcher cannot reach the rectangle, lower it, keep its size, and write the new height on page 4.

Shooting worksheet

Program: Tune Shooter. Shoot five, count, change ONE thing, repeat.

First: on the 34-inch mark, read the camera range 3 times, moving away and back in between: _____ _____ _____ (should be within 1 in). Load one ball and tap Y once: exactly one ball should leave. _____

1. Front bumper on the mark, on the center line. Hold A to face the tags. **2.** Hold right bumper; when At speed, tap Y once per ball. Shoot five. **3.** Write one row. **4.** Mostly low: speed up one step. Mostly high: speed down one step. Left or right: aim again, do not change speed. **5.** At four or more hits, shoot five more to check, then circle the row. **6.** Do the 34, 26 and 18 inch marks.

Rectangle bottom edge: _____ in Target printed: _____ Top speed allowed: _____ Battery re-checked: _____ (SWAP: change it, then re-shoot one row)

Mark (in from wall)	Camera range (in)	Speed (ticks/s)	Hits out of 5	Misses: low, high, left, right	Next change

Put the circled rows in the code

In ShooterCalibration.java, type the circled rows into RANGE_INCHES and SPEED_TICKS_PER_SECOND, closest range first. Between two rows the robot picks a speed in between. Outside the rows it refuses to shoot.

Row 1: range _____ speed _____ Row 2: range _____ speed _____ Row 3: range _____ speed _____

Auto path worksheet

Program: Autonomous Route. Take the balls out; dry run stays on.

A route is a list of steps in Routes.java: **drive(inches)**, **turn(degrees)** and **aimAndShoot()**, which faces the tags, shoots if switched on, and turns back. Start with the back bumper on the start line, on the center line, facing the wall. Type your numbers into the RED_AUDIENCE_A list: first drive, aimAndShoot, park turn, park drive. Change one number per run. Three runs in a row that end in the box means done.

Run	First drive (in)	Range at stop (in)	Park turn (degrees)	Park drive (in)	Ended in box?	Seconds	Next change
1							
2							
3							
4							
5							

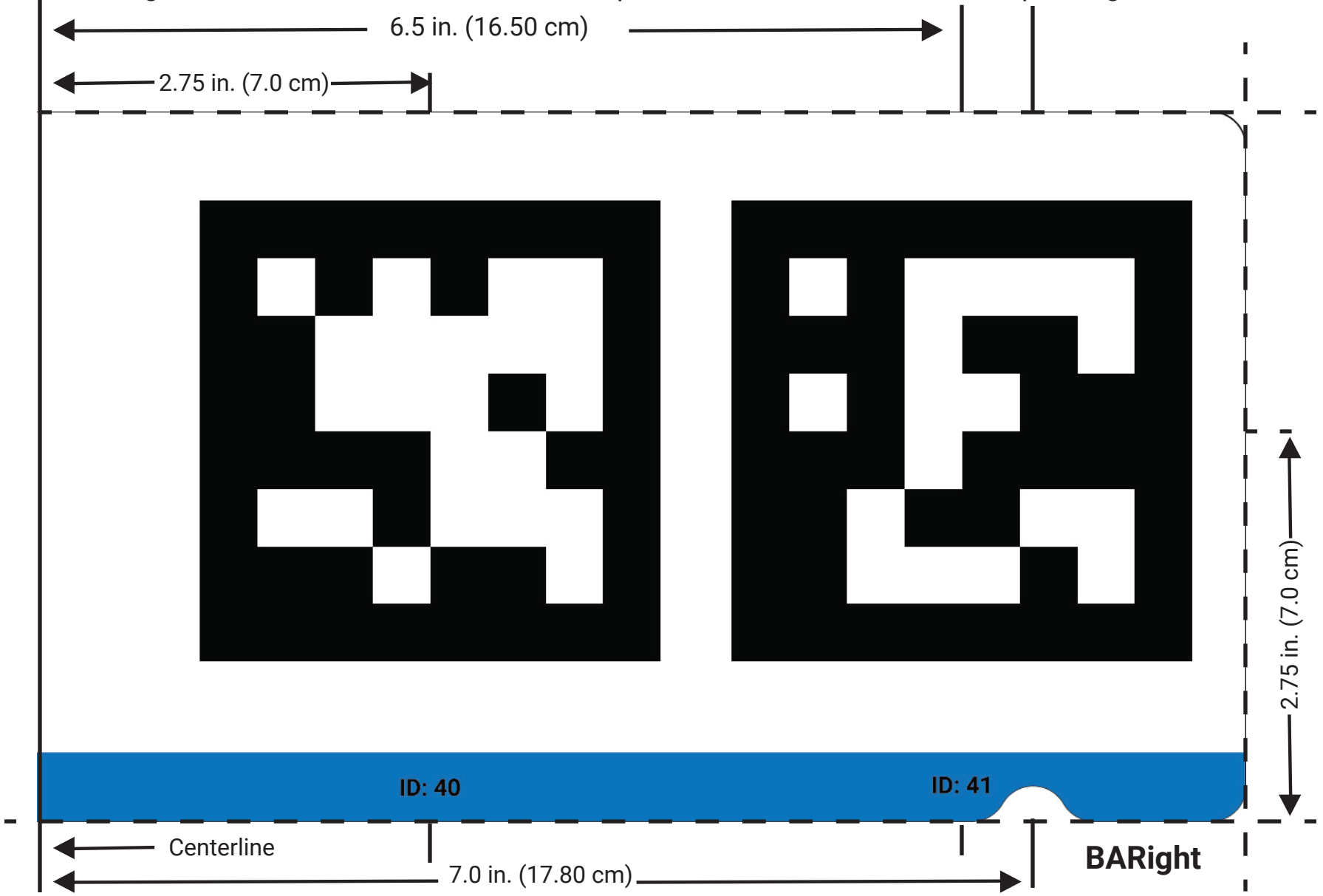
If it stops early

The screen says	Try this
Not moving: ...	Ask the mentor: a setup switch is still off.
Shot skipped: tags not lined up in time	Check camera tilt, lighting, and the target you chose.
Shot skipped: speed table has no rows ...	The first drive stopped outside your measured ranges.
Drive/turn timeout	Something blocked a wheel, or the distance is too long for 4 seconds.

In a match: leaving the field wall in AUTO is 3 points, ending AUTO partly in the LOADING ZONE is 5, and tipping the hive is 20.

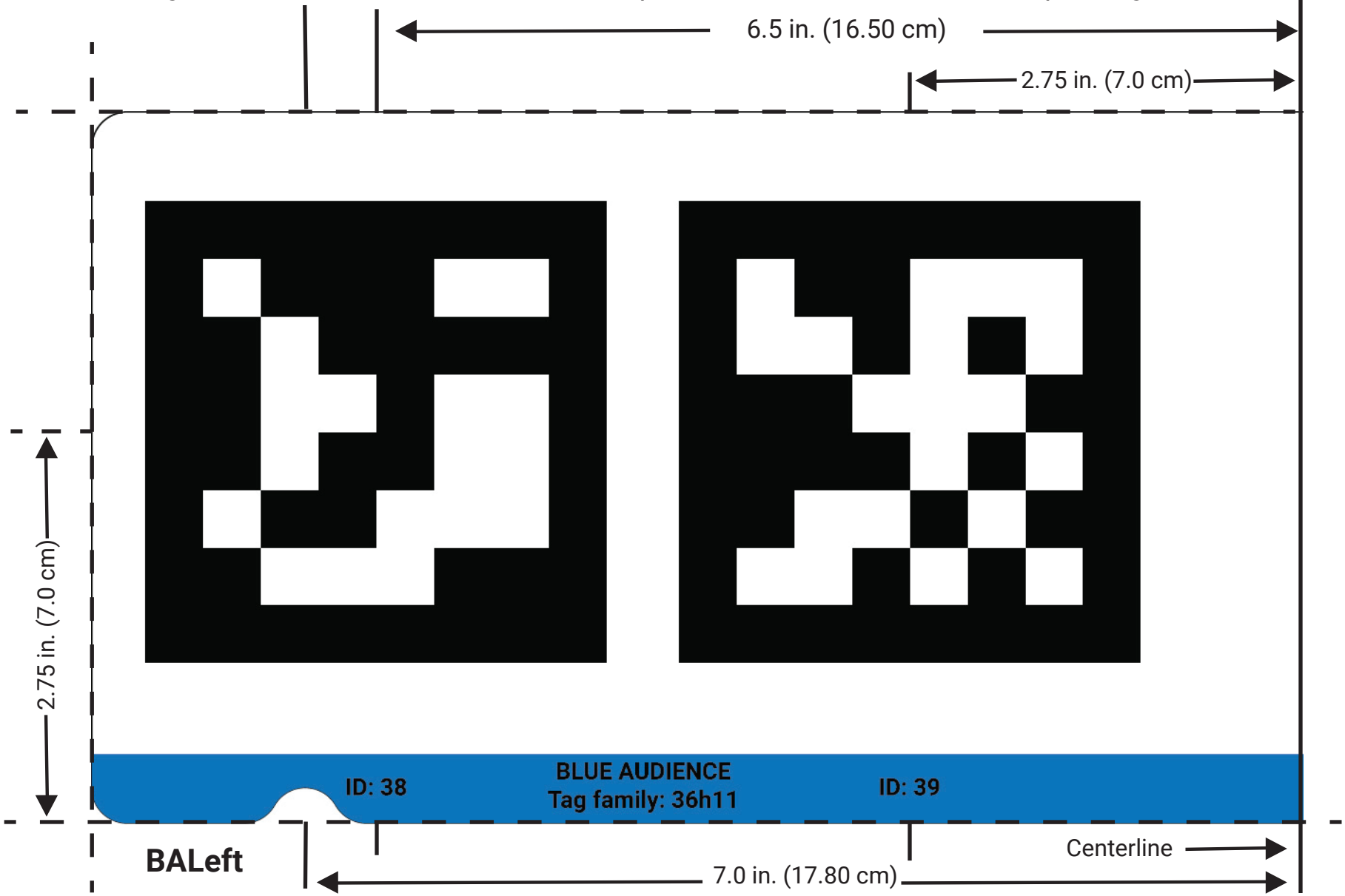
INSTRUCTIONS FOR USE (V1):

1. Print document using ACTUAL SIZE settings or use a “CUSTOM SCALE” at 100%. To verify the document, the measurements indicated should match when printed.
2. Align left and right AprilTag pair at the centerline.
3. Align the two reference holes in the clear plastic of the hive with the corresponding circles.



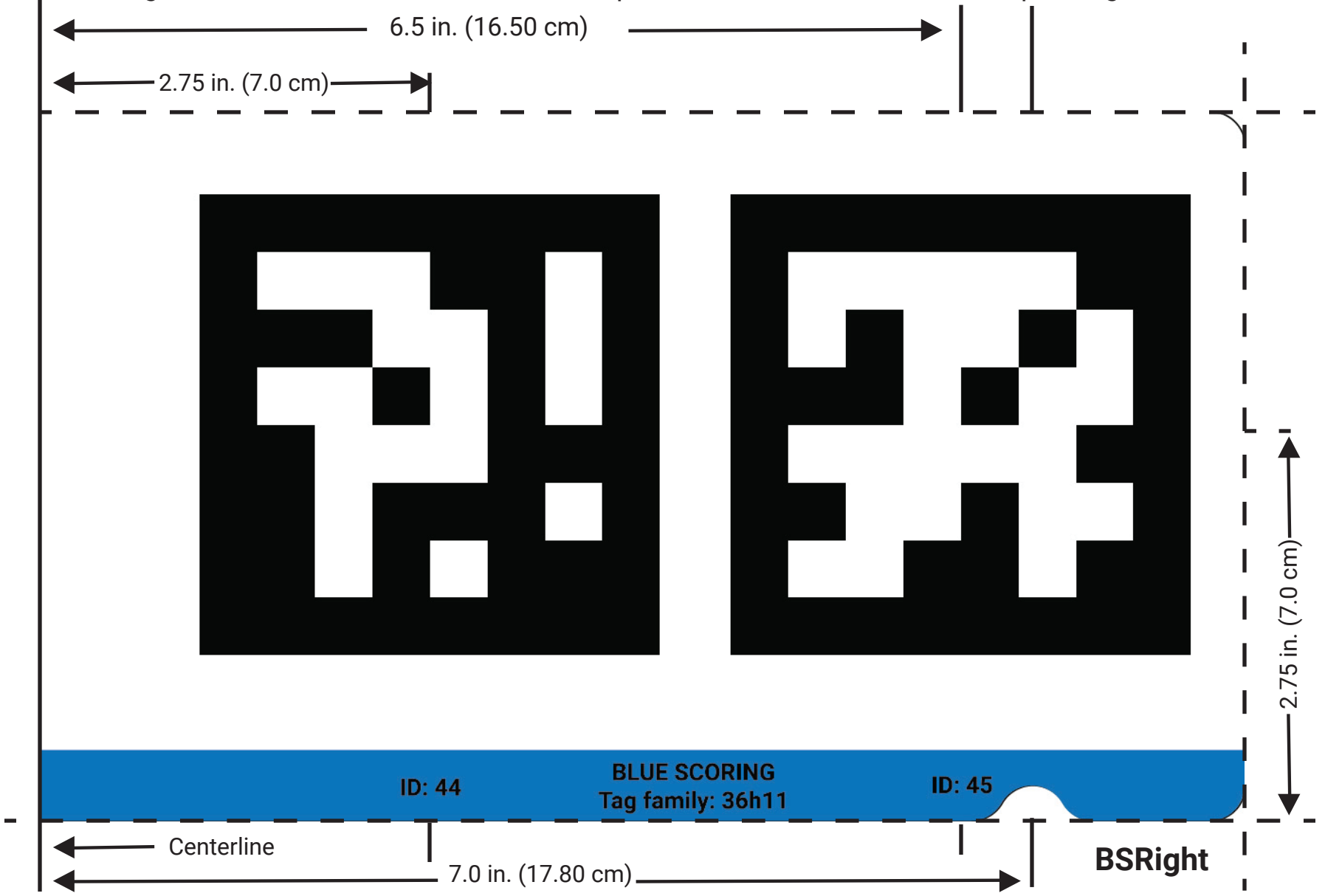
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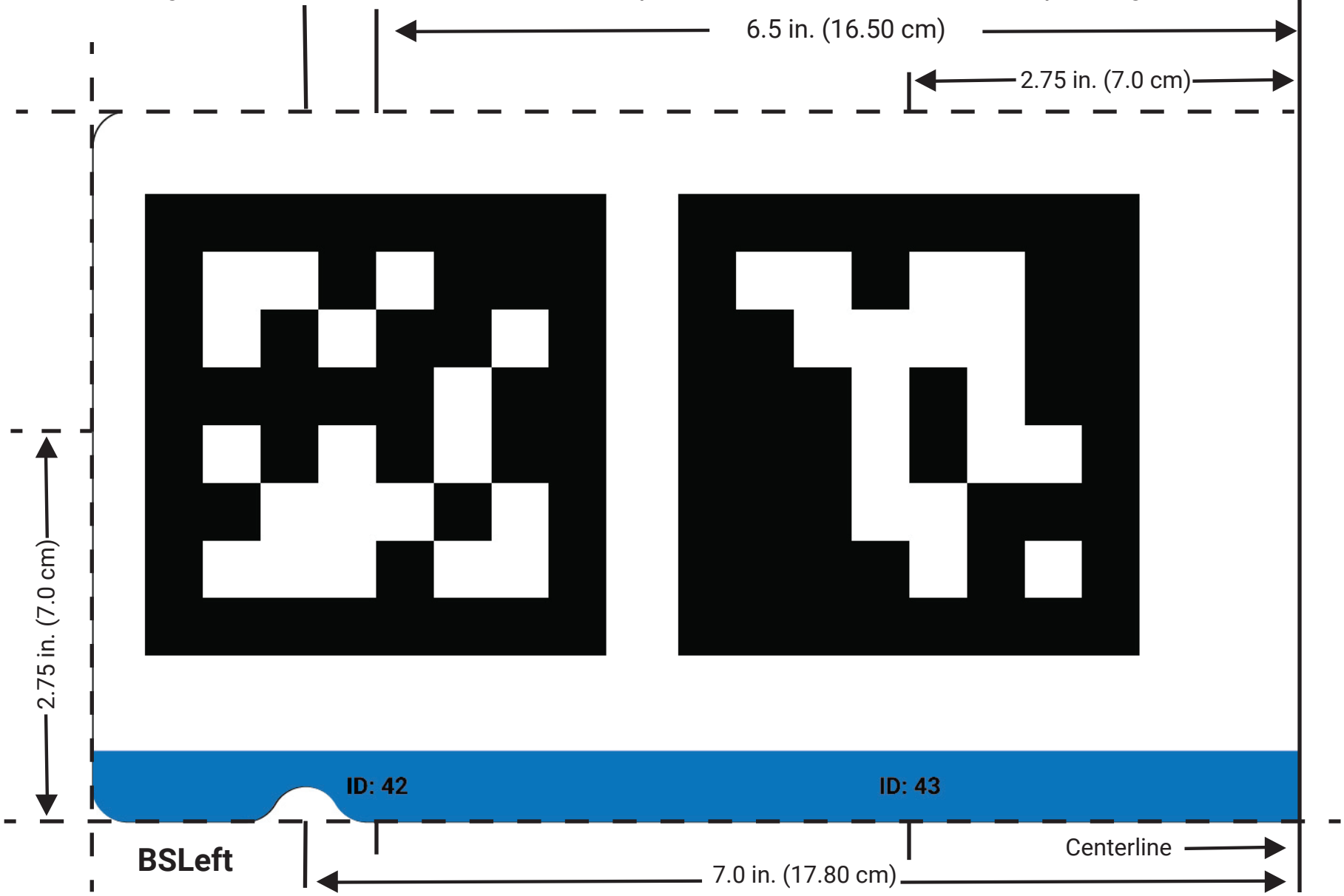
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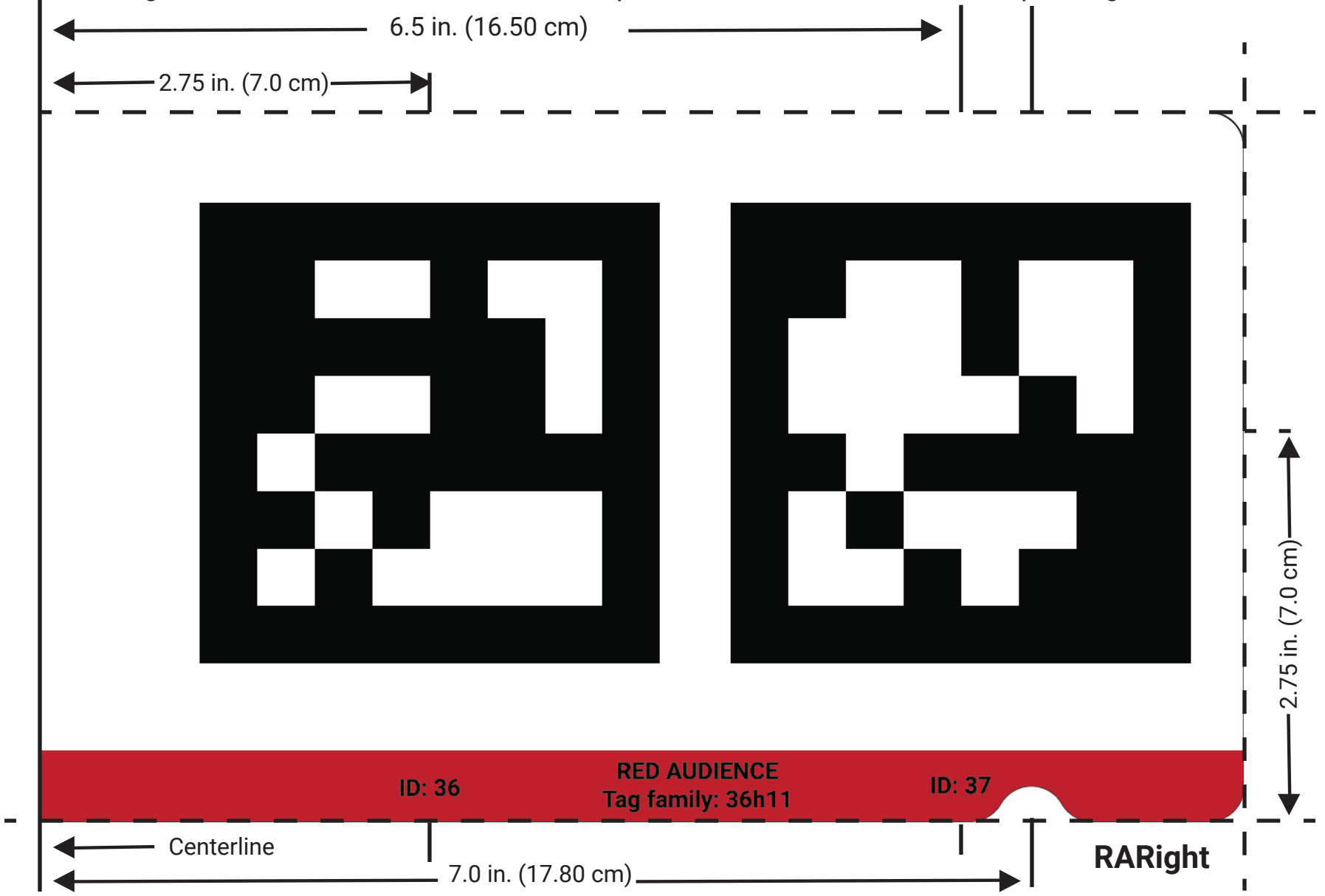
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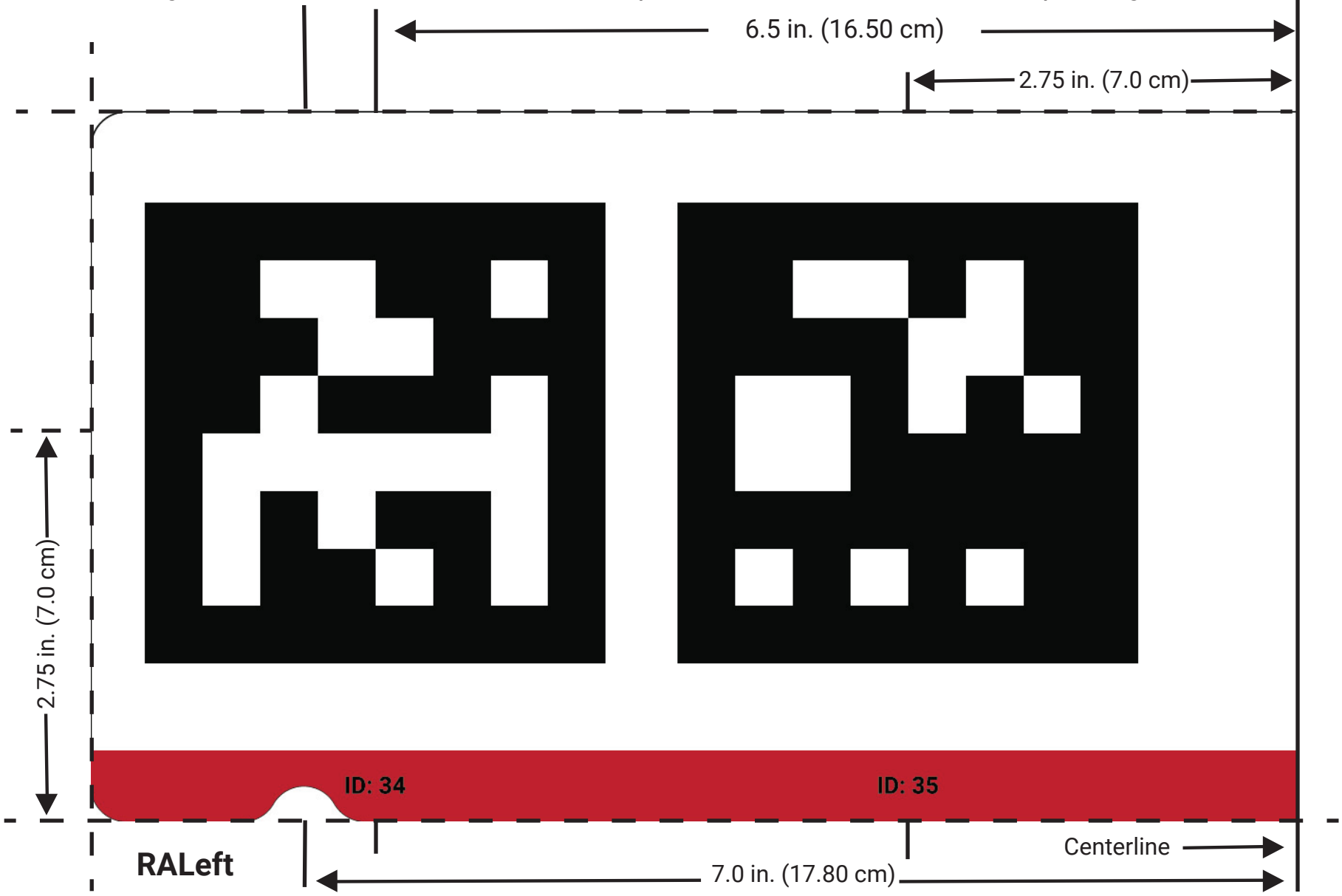
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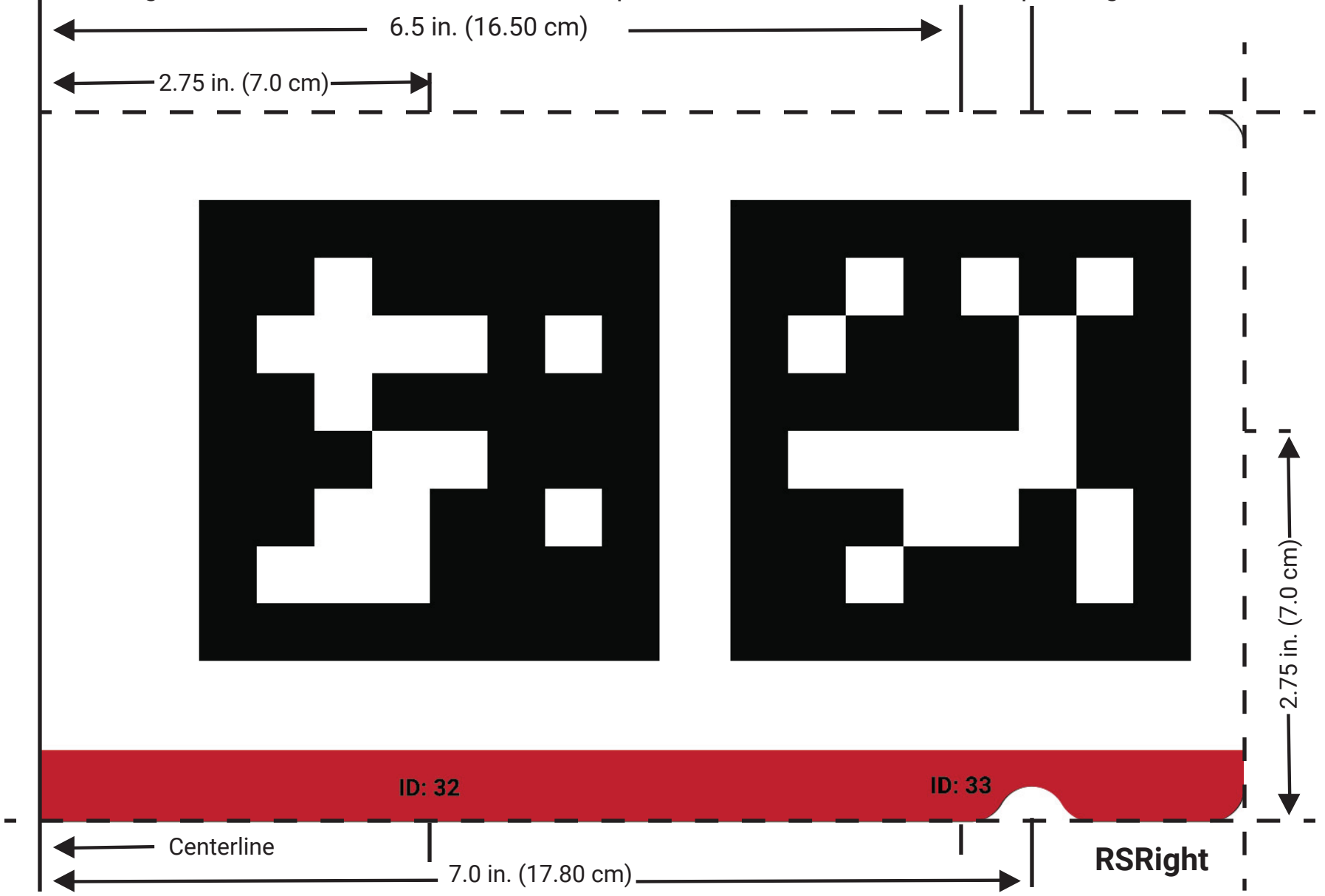
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